

Unit-1

Lecture-4

Photoelectric Effect

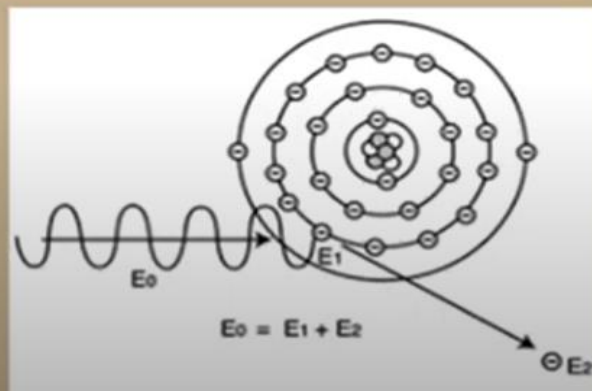
Property Used
to Compare

Compton Effect

Electrons ejected
out of atoms.
Photon is absorbed

Meaning

Electrons are not ejected
out of atoms. Incident
photon is absorbed. A
new photon of lower
frequency is emitted



In the photoelectric
effect, the entire energy
of the incident photon is
acquired by a single
electron

Energy

In the Compton effect,
the incident photon
transfers only a part of
its energy to an electron

Photoelectric Effect	Property Used to Compare	Compton Effect
The photoelectric effect is a low energy phenomenon, and the interacted photons disappear just after they deliver their energy to electrons.	Phenomenon	The Compton effect, on the other hand, is a mid-energy phenomenon, and the interacted photons are scattered by the electrons
The photon delivers its total amount of energy to a single electron.	Photon Energy	The photon transfers part of its energy to a single electron.
Photoelectric effect was explained by Albert Einstein.	Explained by	Compton effect was explained by Arthur Compton.

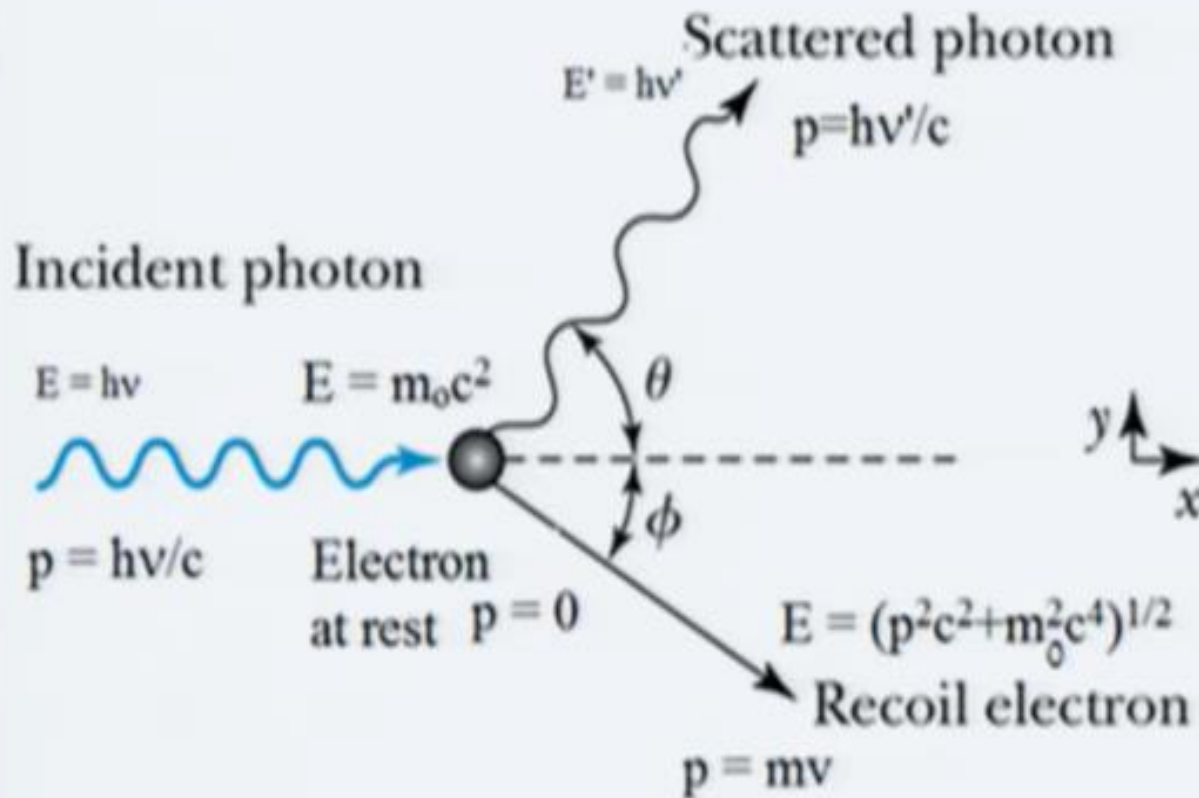
Quantum Explanation of Compton Effect

As per explanation given by Compton, according to **Quantum Theory**, when photon of energy $h\nu$ strikes with the substance, some of the energy of photon is transferred to the electrons, so the energy of photon reduces and **wavelength increases**.

Various assumptions were made for explaining this phenomena as:

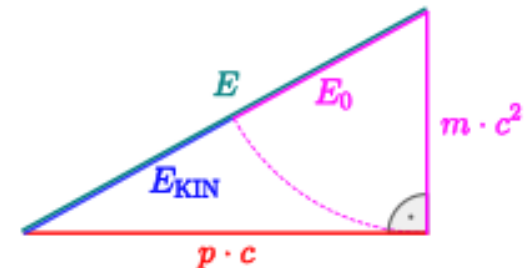
- Compton Effect is the result of interaction of an **individual particle and free electron** of target.
- The collision is **relativistic and elastic**.
- The laws of **conservation** of energy and momentum hold good.

Derivation of Compton Shift



The photon incident on a free electron and scattered. The target electron also recoils as shown figure.

The energy–momentum relation is consistent with the familiar [mass–energy relation](#) in both its interpretations: $E = mc^2$ relates total energy E to the (total) [relativistic mass](#) m , while $E_0 = m_0c^2$ relates [rest energy](#) E_0 to (invariant) rest mass m_0



Derivation of Compton Shift Cont..

The energy of the system before collision = $h\nu + m_0c^2$

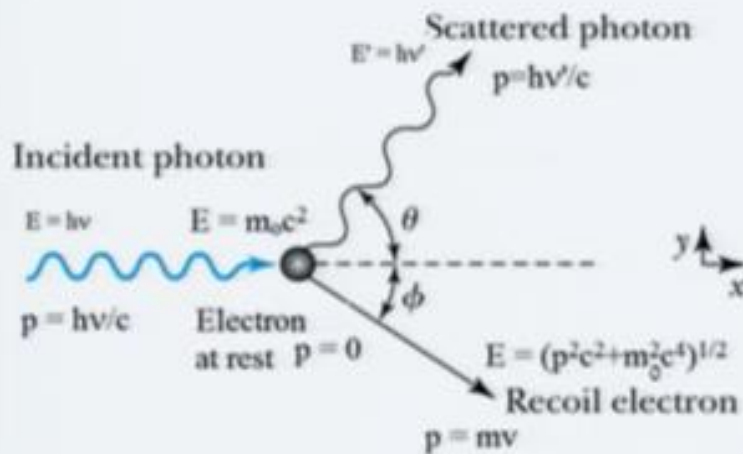
The energy of the system after collision = $h\nu' + (p^2c^2 + m_0^2c^4)^{1/2}$

According to the principle of conservation of energy

$$h\nu + m_0c^2 = h\nu' + \sqrt{p^2c^2 + m_0^2c^4} \quad \text{or} \quad \left[\sqrt{p^2c^2 + m_0^2c^4} \right]^2 = [(h\nu - h\nu') + m_0c^2]^2$$

So

$$p^2c^2 = (h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu') + 2(h\nu - h\nu')m_0c^2 \quad \dots (1)$$



Derivation of Compton Shift Cont..

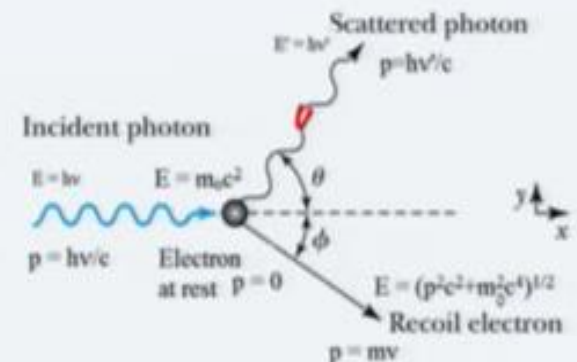
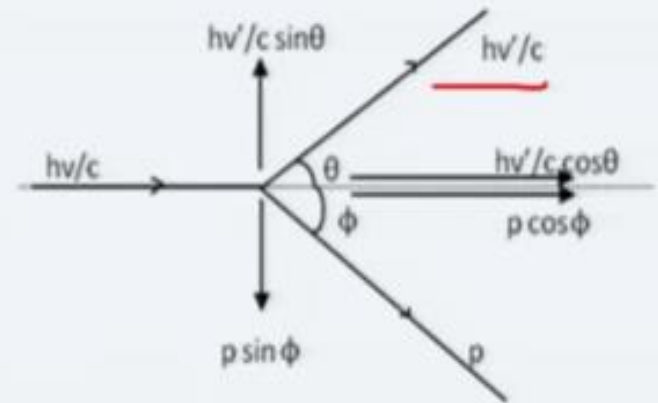
According to law of conservation of momentum

On X axis
$$\frac{h\nu}{c} + 0 = \frac{h\nu'}{c} \cos \theta + p \cos \phi$$

or
$$pc \cos \phi = h\nu - h\nu' \cos \theta \quad \dots (2)$$

On Y axis
$$0 = \frac{h\nu'}{c} \sin \theta - p \sin \phi$$

or
$$pc \sin \phi = h\nu' \sin \theta \quad \dots (3)$$



Derivation of Compton Shift Cont..

From eq. (2) and (3)

$$p^2 c^2 (\sin^2 \phi + \cos^2 \phi) = (h\nu' \sin \theta)^2 + (h\nu - h\nu' \cos \theta)^2$$

$$pc \cos \phi = h\nu - h\nu' \cos \theta \text{ .. (2)}$$

$$pc \sin \phi = h\nu' \sin \theta \text{ .. (3)}$$

$$p^2 c^2 = (h\nu' \sin \theta)^2 + (h\nu)^2 - 2(h\nu)(h\nu' \cos \theta) + (h\nu')^2 \cos^2 \theta$$

$$\text{i.e. } p^2 c^2 = (h\nu)^2 + (h\nu')^2 (\sin^2 \theta + \cos^2 \theta) - 2(h\nu)(h\nu')(\cos \theta)$$

$$p^2 c^2 = (h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu')(\cos \theta) \text{ (4)}$$

$$p^2 c^2 = (h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu') + 2(h\nu - h\nu')m_o c^2 \text{ (1)}$$

Now from eq. (1) and (4)

$$(h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu') + 2(h\nu - h\nu')m_o c^2 =$$

$$(h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu')(\cos \theta)$$

Derivation of Compton Shift Cont..

i.e. $(h\nu - h\nu')m_0c^2 = (h\nu)(h\nu')(1 - \cos \theta)$

so $(\nu - \nu')m_0c^2 = h(\nu\nu')(1 - \cos \theta)$

i.e. $\frac{(\nu - \nu')}{\nu\nu'} = \frac{h}{m_0c^2}(1 - \cos \theta)$

or $\frac{1}{\nu'} - \frac{1}{\nu} = \frac{h}{m_0c^2}(1 - \cos \theta)$ by using $\nu = \frac{c}{\lambda}$

$$\frac{\lambda'}{c} - \frac{\lambda}{c} = \frac{h}{m_0c^2}(1 - \cos \theta)$$

Derivation of Compton Shift Cont..

So we have the expression of Compton shift

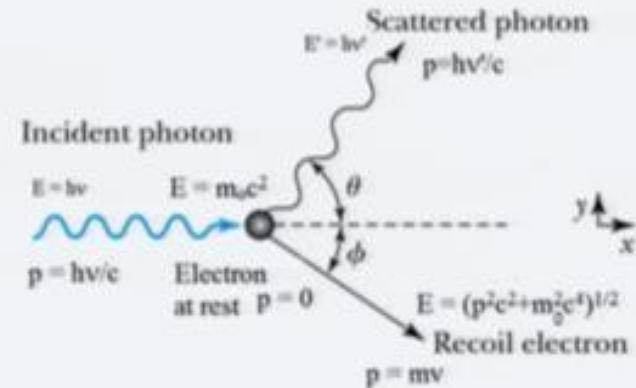
$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_o c} (1 - \cos \theta) \dots (5)$$

It is clear from equation (5) that Compton shift ($\Delta\lambda$) depends only on the angle of scattering and is independent of the wavelength of incident photon.

Here Compton wavelength $\lambda_c = \frac{h}{m_o c}$

Compton shift Discussion

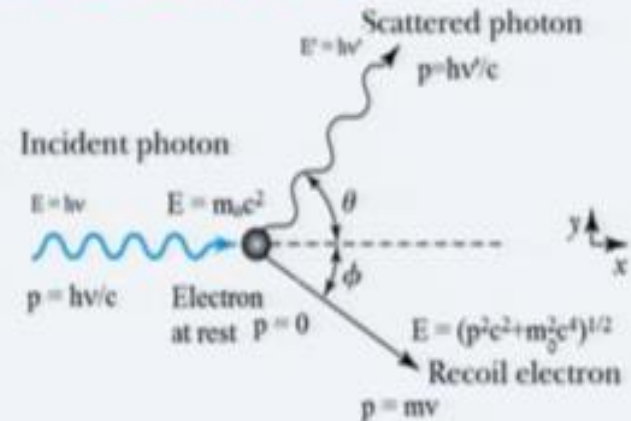
➤ The wavelength $\lambda' = \lambda + \Delta\lambda$ of scattered photon is greater than wavelength λ of incident photon.



➤ When $\theta = 0$, i.e. same as incident direction then $\lambda' - \lambda = \Delta\lambda = 0$, i.e. $\lambda' - \lambda$ means no scattering, which is the case of unmodified scattering.

Compton shift Discussion Cont..

➤ When $\theta = \pi/2$, $\Delta\lambda = \frac{h}{m_0 c} = \lambda_c = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8}$
 $= 2.42 \times 10^{-12} \text{ m} = 0.0242 \text{ \AA}$



Here $\lambda_c = \frac{h}{m_0 c} = 0.0242 \text{ \AA}$ is known as Compton wavelength which is constant.

➤ When $\theta = \pi$, $\Delta\lambda = \frac{2h}{m_0 c} = 2\lambda_c = \frac{2 \times 6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} = 0.0484 \text{ \AA}$

Which is the head-on scattering and represents the maximum change in wavelength.

Numerical Problem

Problem: X- rays with wavelength $1\text{\AA}$ are scattered from a carbon block, the scattered radiation is viewed at 90° to the incident beam. Find the Compton shift $\Delta\lambda$ and wavelength of scattered photon.

Solution: The Compton shift is given by

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_o c} (1 - \cos \theta)$$

$$\Delta\lambda = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} (1 - \cos 90^\circ) = 0.0242 \text{\AA}$$

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_o c} (1 - \cos \theta)$$

$$\underline{\lambda'} = \lambda + \frac{h}{m_o c} (1 - \cos \theta)$$

$$\lambda' = 1 \times 10^{-10} \text{\AA} + \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} (1 - \cos 90^\circ) \text{\AA}$$

$$\lambda' = 1 \times 10^{-10} \text{\AA} + 2.42 \times 10^{-12} \text{\AA} = 1.0242 \text{\AA}$$

$$\lambda' = 1.0242 \text{\AA}$$